

Lifestyle & Health: Hormones

Answer sheet · 37 marks total

Section A — The endocrine system & hormones vs nerves

- A1.** The system of glands that secrete hormones (1) directly into the bloodstream (1).
- A2.** A chemical messenger (1) carried in the blood to a target organ, where it produces an effect (1).
- A3.** Any two: nervous system uses electrical impulses along neurons, hormonal system uses chemicals in the blood (1); nervous responses are fast, hormonal responses are slower (1); nervous effects are short-lived/precise, hormonal effects are longer-lasting/widespread (1).

Section B — Blood glucose regulation

- B1.** Pancreas (1).
- B2.** The pancreas detects the rise and releases insulin (1); insulin causes cells (e.g. liver/muscle) to take up glucose from the blood (1); blood glucose concentration falls back to normal (1).
- B3.** It is converted into glycogen (1) and stored, e.g. in the liver/muscles (1).

Section C — Type 1 & Type 2 diabetes

- C1.** The pancreas produces little or no insulin (1), so blood glucose cannot be properly controlled (1).
- C2.** Body cells stop responding properly to insulin/become resistant to it (1), or the pancreas does not produce enough insulin (1).
- C3.** Regular insulin injections (1), matched to diet and activity level, with blood glucose monitoring (1).
- C4.** Lifestyle changes such as a healthier diet, more exercise and weight loss (1); medication and/or insulin injections if needed (1).
- C5.** Blood glucose concentration stays too high for long periods (1); this can damage blood vessels and nerves, increasing the risk of further disease, e.g. cardiovascular disease (1).

Section D — Hormones, lifestyle & health risk

- D1.** Carrying excess body fat makes cells less responsive to insulin/causes insulin resistance (1), so blood glucose is not controlled properly, increasing the risk of Type 2 diabetes (1).
- D2.** Any two: poor diet, e.g. high in fat/sugar (1); lack of exercise (1); (accept family history/genetics with justification) (1).
- D3.** Type 1 diabetes is caused by the pancreas producing little or no insulin (often an autoimmune response), not by lifestyle factors (1); Type 2 diabetes is strongly linked to lifestyle factors such as obesity, poor diet and inactivity, so lifestyle changes reduce the risk (1).

Section E — Apply it

- E1.** Patient X (Type 1): will likely need regular insulin injections (1) alongside monitoring blood glucose and matching insulin to diet/activity (1). Patient Y (Type 2): will likely be advised to make lifestyle changes first, e.g. improved diet, more exercise, weight loss (1), with medication and/or insulin added later if these are not enough to control blood glucose (1).
- E2.** The statement is incorrect because insulin causes blood glucose concentration to fall, not rise (1); insulin does this by causing cells to take up glucose from the blood and by increasing conversion of glucose to glycogen for storage (1); it is glucagon, not insulin, that causes blood glucose to rise (1).

E3. It shows whether blood glucose is being kept within a safe range (1), allowing insulin dose, diet or activity to be adjusted to avoid blood glucose being too high or too low (1).

Marking note: accept any scientifically correct alternative answer with the same reasoning, even if different from the example shown — award marks for a correctly linked point, not for matching these exact words.